

← Attempt 1

Date Started

September 10, 2026 9:28 pm

Date Completed

September 10, 2026 9:28 pm

Time Spent

0m 5s

Score

Waiting on Instructor

☐ Hide correct responses

Question 1.1

A signed number (regardless of representation) can be thought of as a vector with an origin at

1	
---	--

, magnitude equal to the number's

2	
---	--

, and direction indicated by the number's

3	
---	--

.

Correct answers:

1 zero 2 absolute value 3 sign

 Attempt 1 of 3 Score: 0/0 pts

Question 1.2

When representing a value in **N-bit signed-magnitude format**, the

1	
---	--

 bit represents the value's

2	
---	--

, and the remaining

3	
---	--

 bits are the binary representation of the value's

4	
---	--

.

Correct answers:

1 most-significant 2 sign 3 N - 1 4 absolute value

 Attempt 1 of 3 Score: 0/0 pts

Question 1.3

Which of the following types of numbers can be represented in **unsigned** binary format?

	Unsigned Can Represent	Unsigned Can NOT Represent
Negative	☐	☒ ✓
Zero	☒ ✓	☐
Positive	☒ ✓	☐

Attempt 1 of 1 Score: 0/0 pts

Question 1.4

Which of the following types of numbers can be represented in **signed-magnitude** binary format?

	Signed-Magnitude Can Represent	Signed-Magnitude Can NOT Represent
Negative	☒ ✓	☐
Zero	☒ ✓	☐
Positive	☒ ✓	☐

Attempt 1 of 1 Score: 0/0 pts

Question 1.5

Which of the following types of numbers can be represented in **2's-complement** binary format?

	2's-Complement Can Represent	2's-Complement Can NOT Represent
Negative	☐ ✓	☐
Zero	☐ ✓	☐
Positive	☐ ✓	☐

Attempt 1 of 1 Score: 0/0 pts

Question 1.6

For both signed-magnitude and 2's-complement formats, a in the most-significant bit position indicates the number is negative, whereas a in that position indicates it is positive (or zero).

Correct answers:

1 1 2 0

Attempt 1 of 1 Score: 0/0 pts

Question 2.1

To **negate** a number means to

1	
---	--

.

Correct answers:

- 1 give it the opposite sign (multiply by -1)

Attempt 1 of 1 Score: 0/0 pts

If we negate +42, we get -42.

If we negate -42, we get +42.

Question 2.2

To negate a 2's-complement number, we

1	
---	--

, then

2	
---	--

.

Correct answers:

- 1 complement all of the bits
- 2 add 1

Attempt 1 of 2 Score: 0/0 pts

Question 2.3

The above procedure works for

1	
---	--

.

Correct answers:

- 1 negating either positive or negative numbers

Attempt 1 of 2 Score: 0/0 pts

Question 2.4

When we convert a **negative** decimal number to 2's-complement format, we first determine the binary representation of its absolute value, then

1	
---	--

 that value.

Correct answers:

- 1 negate

Attempt 1 of 2 Score: 0/0 pts

Question 2.5

When we convert a **positive** number to 2's-complement format, we

1	
---	--

 need to complement all the bits and add 1.

Correct answers:

1 DO NOT

Attempt 1 of 1 Score: 0/0 pts

Complementing all the bits and adding 1 negates the number.

If you start with a positive number and negate it, you get a negative number, which has a different **value**!

Converting a number to a different format should **NOT** change its **value**, only **how that value is represented**.

Question 2.6

Remember, we know that a 2's-complement number is negative if it has a

1	
---	--

 in the most-significant (leftmost) position.

Correct answers:

1 1

Attempt 1 of 1 Score: 0/0 pts

Question 2.7

To convert a 2's-complement number to decimal, we:

1. Take the absolute value of the number by if it is negative, or if it is positive.
2. Convert the resulting binary number to decimal
3. Write down a minus sign in front of the decimal number .

Correct answers:

- 1 negating it 2 doing nothing to it 3 positive
- 4 only if the original 2's-complement number was negative

Attempt 1 of 1 Score: 0/0 pts

Question 3.1

Given the binary number **000...**

...if it is an **unsigned** binary number, its **value** (in decimal) is

...if it is a **signed-magnitude** binary number, its **value** (in decimal) is

...if it is a **2's-complement** binary number, its **value** (in decimal) is

Correct answers:

- 1 0 2 0 3 0

Attempt 1 of 3 Score: 0/0 pts

Question 3.2

Given the binary number **111**...

...if it is an **unsigned** binary number, its **value** (in decimal) is

1	
---	--

...if it is a **signed-magnitude** binary number, its **value** (in decimal) is

2	
---	--

...if it is a **2's-complement** binary number, its **value** (in decimal) is

3	
---	--

Correct answers:

1 7 2 -3 3 -1

Attempt 1 of 3 Score: 0/0 pts

Question 3.3

Given the binary number **100**...

...if it is an **unsigned** binary number, its **value** (in decimal) is

1	
---	--

...if it is a **signed-magnitude** binary number, its **value** (in decimal) is

2	
---	--

...if it is a **2's-complement** binary number, its **value** (in decimal) is

3	
---	--

Correct answers:

1 4 2 0 3 -4

Attempt 1 of 3 Score: 0/0 pts

Question 3.4

Given the binary number **010...**

...if it is an **unsigned** binary number, its **value** (in decimal) is

1	
---	--

...if it is a **signed-magnitude** binary number, its **value** (in decimal) is

2	
---	--

...if it is a **2's-complement** binary number, its **value** (in decimal) is

3	
---	--

Correct answers:

1 2 2 2 3 2

Attempt 1 of 3 Score: 0/0 pts

Question 4

The below image shows a **number line** labeled in decimal, with three rows above representing different 3-bit binary formats.

For each box indicated on the number line, drag-and-drop the number that represents the required value (equal to the decimal number at that position) in the indicated format for that row.

If a value cannot be represented in the indicated format, mark it with the **X**.

Format	Box 1 (-3)	Box 2 (-2)	Box 3 (-1)	Box 4 (0)	Box 5 (1)	Box 6 (2)	Box 7 (3)	Box 8 (4)	Box 9 (5)	Box 10 (6)	Box 11 (7)	Box 12 (8)
Signed-magnitude	1	3	6						9			
2's-comp	2	4	7						11			
Unsigned		5	8					10			12	

Drag-and-drop options: X, 000, 001, 010, 011, 100, 101, 110, 111

Correct answers:

1 111 2 101 3 101 4 111 5 X 6 001 7 011 8 011
 9 X 10 101 11 X 12 111

Attempt 1 of 3 Score: 0/0 pts

Notice that the two different signed number formats represent the same negative values using different binary numbers!

Positive numbers, however, look the same in all formats (if they are representable in the given number of bits).

Question 5.1

Imagine that all possible 3-bit **signed-magnitude** binary numbers are arranged on a number line.

The **number** furthest to the left on the number line would be:

1	
---	--

The **value** of this number (in decimal) is:

2	
---	--

.

Correct answers:

1 111 2 -3

Attempt 1 of 3 Score: 0/0 pts

Question 5.2

For the same 3-bit **signed-magnitude** number line described in the previous question...

The **number** furthest to the right on the number line would be:

1	
---	--

The **value** of this number (in decimal) is:

2	
---	--

Correct answers:

1 011 2 3

Attempt 1 of 3 Score: 0/0 pts

Question 5.3

Now imagine that all possible 3-bit **2's-complement** numbers are arranged on a number line.

The **number** furthest to the left on the number line would be:

1	
---	--

The **value** of this number (in decimal) is:

2	
---	--

Correct answers:

1 100 2 -4

Attempt 1 of 3 Score: 0/0 pts

Question 5.4

For the same 3-bit **2's-complement** number line described in the previous question...

The **number** furthest to the right on the number line would be:

1	
---	--

The **value** of this number (in decimal) is:

2	
---	--

Correct answers:

1 011 2 3

Attempt 1 of 3 Score: 0/0 pts

Question 6.1

Convert the 4-bit unsigned binary number 0010 to **4-bit 2's-complement** format.

Correct answer:

0010

Attempt 1 of 3 Score: 0/0 pts

Question 6.2

Convert the decimal number -7 to **4-bit 2's-complement** format.

Correct answer:

1001

Attempt 1 of 3 Score: 0/0 pts

Question 7.1

We use

1	
---	--

 to transform an unsigned binary number into another that uses more bits, but has the same value.

This process involves inserting

2	
---	--

 as many times as needed at the

3	
---	--

 end of the number.

Correct answers:

- 1 zero extension 2 a zero 3 most-significant

Attempt 1 of 2 Score: 0/0 pts

Question 7.2

We use **sign extension** to transform a 2's-complement number into another that

1	
---	--

.

If we perform sign extension correctly, the value of the new number

2	
---	--

 be equal to the value of original number.

Correct answers:

- 1 uses more bits 2 will

Attempt 1 of 2 Score: 0/0 pts

Question 7.3

To sign-extend a 2's-complement number, we insert

1	
---	--

 as many times as needed at the

2	
---	--

 end of the number.

Correct answers:

- 1 the value of the leftmost bit 2 most-significant

Attempt 1 of 2 Score: 0/0 pts

Sign-extension does NOT change the value of a number – it just increases the number of bits being used to represent that value!

Question 8.1

What does it mean if a binary addition (either signed or unsigned) results in **overflow**?

- ☐ Adding the column of bits in the most significant position resulted in a carry.
- ☐ The result is larger than the largest operand.
- ☒ The result cannot be correctly represented in the required number of bits. ✓
- ☐ The summation resulted in a carry value equal to 1 in at least one of the positions.

Attempt 1 of 2 Score: 0/0 pts

Question 8.2

If adding two **negative** 2's-complement numbers **overflows**, the sign of the result will appear to be

1	
---	--

.

Hint: if you're not sure, try a couple of examples...

Correct answers:

1 non-negative

Attempt 1 of 1 Score: 0/0 pts

Question 8.3

If the result of adding two **positive** 2's-complement numbers appears to be **positive**, then the operation

1	
---	--

.

Hint: if you're not sure, try a couple of examples...

Correct answers:

1 did not overflow

Attempt 1 of 1 Score: 0/0 pts

Question 8.4

If adding two **unsigned** numbers **overflows**, the result will appear to be

1	
---	--

.

Hint: if you're not sure, try a couple of examples...

Correct answers:

- 1 smaller than it should be

Attempt 1 of 1 Score: 0/0 pts

Question 8.5

We can determine if a particular (correctly-calculated) addition caused overflow using one common technique regardless of whether the operands are unsigned, positive, or negative by asking the question:

1	
---	--

. If so, the operation overflowed.

Correct answers:

- 1 "Is the result incorrect?"

Attempt 1 of 1 Score: 0/0 pts

Overflow occurs if the result does not have enough bits to represent the correct value, so the **result is wrong**.

Question 9.1

What 4-bit binary number should you add to the number shown to get a 4-bit result of 0 (zero)?

$$\begin{array}{r} 0111 \\ + ??? \\ \hline 0000 \end{array}$$

Hint: You should be able to figure this out just by working your way from the least-significant bit up to the most-significant bit, choosing the bit value that will make the sum for that column equal 0.

Correct answer:

1001

Attempt 1 of 3 Score: 0/0 pts

Question 9.2

If you take the correct 4-bit answer from the previous question, and interpret it as a **2's-complement number**, what is its value in **decimal**?

Correct answer:

-7

Attempt 1 of 3 Score: 0/0 pts

Question 9.3

What was the value (in **decimal**) of the original number you were given at the top of this page (the value on top in the addition)?

Correct answer:

7

Attempt 1 of 3 Score: 0/0 pts

The whole reason we use the 2's-complement representation in computing is that the math works as it should ($7 + (-7) = 0$).

This means we can use exactly the same hardware for signed addition as we do for unsigned addition!

Question 10.1

Suppose that we need to perform a subtraction operation $A - B$, where A and B are N -bit binary numbers. In digital hardware, rather than using "normal" subtraction with borrowing, we use

1	
---	--

 binary representation and perform the operation

2	
---	--

 +

3	
---	--

.

Correct answers:

1 2's-complement 2 A 3 $(-B)$

Attempt 1 of 2 Score: 0/0 pts

Question 10.2

Suppose that you are performing the subtraction operation $-4_{10} - 2_{10}$ in **binary** with 3-bit operands and are limited to a **3-bit result**.

What result would we calculate for this operation? Answer in decimal.

Correct answers:

2

Attempt 1 of 3 Score: 0/0 pts

Question 10.3

The subtraction operation in the previous problem

1	
---	--

.

We can tell this happened because

2	
---	--

.

Correct answers:

1 overflowed 2 the result was incorrect

Attempt 1 of 2 Score: 0/0 pts